Type-II Fatty-Acid Synthesis in *Pseudomonas putida* KT2440 — Module/ Pathway Curation Review

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Pathway bucket:** KEGG `ppu00061` "Fatty acid biosynthesis" (module area: lipid_cell_envelope_metabolism) **Assignment:** Species-aware satisfiability and annotation-curation review of the dissociated (type-II) bacterial fatty-acid synthase (FAS-II) module. The candidate gene list is treated as a starting point, not ground truth.

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1. Executive Summary

The complete **type-II (dissociated) fatty-acid synthesis pathway is encoded and satisfiable** in *Pseudomonas putida* KT2440. Every canonical FAS-II step — malonyl-CoA supply by the heteromeric acetyl-CoA carboxylase (AccABCD), malonyl-ACP loading by FabD, KAS-III chain initiation, the four-reaction elongation cycle (condensation, β-keto reduction, dehydration, enoyl reduction), and the FabA/FabB *cis*-unsaturation branch — is covered by high-confidence canonical genes present in the genome. From a module-satisfiability standpoint this module should be marked **covered**, subject to a small number of curation actions.

Three findings are the curation focus. **First**, the essential **acyl carrier protein AcpP (PP_1915, Q88LL5)** — the physical substrate carrier on which the entire pathway operates — is present, essential, and supported by protein-level evidence (PE:1), yet is **missing from the 19-gene candidate list**. It sits in the canonical *plsX–fabD–fabG–acpP–fabF* cluster and must be added. **Second**, enoyl reduction is served **solely by FabV (PP_4635, NADH-dependent, EC 1.3.1.9)**: KT2440 encodes no FabI or FabL, so FabV is a single point of coverage for this step, not a redundant paralog. **Third**, ACP activation (holo-ACP synthesis, EC 2.7.8.7) is not performed by a dedicated AcpS; instead a single **bifunctional Sfp/EntD-type 4′-phosphopantetheinyl transferase (PP_1183)** most plausibly activates both AcpP and secondary-metabolite carrier proteins — a *Pseudomonas* lineage substitution rather than a genuine gap.

Consequently, several genes in the candidate bucket are **out-of-scope or over-propagated** annotations: the long-chain acyl-CoA ligases *fadD-I/II* (PP_4549/PP_4550) belong to β-oxidation/fatty-acid activation; PP_2783 is an aromatic (p-cumate) catabolism dehydrogenase; and PP_0581, PP_1852, and PP_2540 are bare short-chain dehydrogenase/reductase (SDR) paralogs with no gene name, EC, or function comment that were swept into the bucket by broad EC/GO mapping. The 84-aa PP_4529 "3-oxoacyl-ACP synthase" is a fragment, not a real condensing enzyme. The chain-initiation gene **PP_4379/FabH** is genuinely present but weakly and inconsistently annotated (labelled "synthase I" while carrying the KAS-III EC number 2.3.1.180) and should be promoted to full review, alongside **PP_1183** (ACP-activating PPTase) and **AcpP**.

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2. Target-Organism Pathway Definition

What the module includes

The module is the **dissociated bacterial FAS-II system**: a set of discrete, monofunctional, freely diffusing enzymes that iteratively build a saturated or *cis*-monounsaturated acyl chain on a small acidic acyl carrier protein (ACP). This is mechanistically and evolutionarily distinct from the **type-I FAS** of animals/fungi (a single large multidomain megasynthase) and from **type-I/type-II polyketide synthases**. In KT2440 the biochemical scope is:

1. **Malonyl-CoA formation** — biotin-dependent carboxylation of acetyl-CoA by the four-subunit AccABCD acetyl-CoA carboxylase.
2. **ACP provision and activation** — apo-AcpP is converted to holo-AcpP by transfer of a 4′-phosphopantetheine arm (PPTase; EC 2.7.8.7).
3. **Malonyl-ACP formation** — FabD transacylates malonate from CoA to holo-ACP.
4. **Chain initiation** — KAS-III (FabH) condenses acetyl-CoA with malonyl-ACP to give acetoacetyl-ACP.
5. **Iterative elongation** — a four-reaction cycle: KAS-I/KAS-II condensation → NADPH β-keto reduction (FabG) → (3R)-hydroxyacyl dehydration (FabZ/FabA) → enoyl reduction (FabV).
6. ***cis*-unsaturation branch** — FabA isomerizes trans-2- to cis-3-decenoyl-ACP; FabB elongates it, committing the chain to the unsaturated series (16:1, 18:1 vaccenate).

Neighboring pathways to keep separate

- **KEGG ppu00071 / β-oxidation (fatty-acid degradation and activation):** the *fadD* long-chain acyl-CoA ligases (PP_4549/PP_4550) activate exogenous fatty acids for catabolism, **not** de novo synthesis. They should not satisfy any FAS-II step.
- **Aromatic-compound catabolism (p-cumate/biphenyl degradation):** PP_2783 is an SDR of this pathway, unrelated to acyl-ACP chemistry.
- **Phospholipid / membrane-lipid assembly (downstream):** PlsX/PlsB/PlsC acyltransferases consume acyl-ACP; they are the sink of FAS-II but not part of the elongation module.
- **Secondary-metabolite / polyketide biosynthesis:** shares the PPTase (PP_1183) and SDR/KS enzyme families, a frequent source of over-propagated annotations.
- **Broad overview maps** (KEGG map01100 "Metabolic pathways", map01212 "Fatty acid metabolism") aggregate synthesis + degradation and should not be used to define this module's boundaries.

Alternate names / database definitions

Type-II FAS; dissociated FAS; FAS-II; KEGG "Fatty acid biosynthesis" (map00061 / ppu00061); MetaCyc "fatty acid biosynthesis initiation/elongation". Condensing enzymes carry the historical "β-ketoacyl-ACP synthase" (KAS I/II/III = FabB/FabF/FabH) nomenclature, which is a recurring source of EC/label confusion (see §5).

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3. Expected Step Model and Satisfiability

#	Step	Activity (EC)	KT2440 gene(s)	Status
1	Malonyl-CoA formation	ACC (6.3.4.14 / 2.1.3.15)	accA PP_1607, accB PP_0559, accC PP_0558, accD PP_1996	**covered**
2a	ACP (carrier)	acyl carrier activity	**acpP PP_1915**	**covered (missing from list)**
2b	ACP activation → holo-ACP	2.7.8.7	**PP_1183 (Sfp/EntD-type)**	**covered by lineage substitution**
3	Malonyl-ACP loading	2.3.1.39	fabD PP_1913	**covered**
4	Chain initiation (KAS-III)	2.3.1.180	**PP_4379 (FabH)**	**covered (weak annotation)**
5a	Condensation (KAS-I)	2.3.1.41	fabB PP_4175	**covered**
5a	Condensation (KAS-II)	2.3.1.179	fabF PP_1916 (+ paralog PP_3303)	**covered**
5b	β-keto reduction	1.1.1.100	fabG PP_1914	**covered**
5c	Dehydration	4.2.1.59	fabZ PP_1602 (+ fabA PP_4174)	**covered**
5d	Enoyl reduction	1.3.1.9 (NADH)	**fabV PP_4635 (sole)**	**covered (single point)**
6	*cis*-unsaturation branch	5.3.3.14 + 2.3.1.41	fabA PP_4174 + fabB PP_4175	**covered**

Every expected step is satisfiable. Two entries require curator attention because the responsible gene is either missing from the metadata (AcpP) or provided by a lineage-specific alternative (PP_1183 for ACP activation; FabV as sole enoyl reductase).

4. Candidate Genes and Evidence

High-confidence core genes (covered)

Acetyl-CoA carboxylase — accA/B/C/D (PP_1607, PP_0559, PP_0558, PP_1996). UniProt annotates the full heteromeric ACC: biotin carboxylase AccC (EC 6.3.4.14), biotin carboxyl carrier AccB, and the carboxyltransferase α/β subunits AccA/AccD (EC 2.1.3.15), with PATHWAY "malonyl-CoA biosynthesis; step 1/1". This is the committed entry into FAS-II and is fully covered (Finding F005).

FabD (PP_1913, EC 2.3.1.39). Malonyl-CoA:ACP transacylase, fabD family — canonical, unambiguous (F005). *Curation note:* the metadata assigns FabD to bucket `ppu00074`, but its function is squarely FAS-II; the bucket label is a database artifact, not a functional discrepancy.

FabG (PP_1914, EC 1.1.1.100). NADPH-dependent 3-oxoacyl-ACP reductase, located immediately upstream of *acpP* in the fab cluster — the canonical, essential β-keto reductase (F005). PP_0581 and PP_2540 (below) are **not** substitutes.

FabZ (PP_1602, EC 4.2.1.59). (3R)-hydroxyacyl-ACP dehydratase, the general elongation-cycle dehydratase; covered (F005).

FabF (PP_1916, KAS-II, EC 2.3.1.179) and FabB (PP_4175, KAS-I, EC 2.3.1.41). The two elongation condensing enzymes. FabF/PP_3303 carry the classic 16:1→18:1 *cis*-vaccenate thermal-regulation role in their UniProt FUNCTION text. **PP_3303 (EC 2.3.1.179)** is a genuine second KAS-II paralog (F005/F006) — a real duplication in *Pseudomonas*, not spurious. Curation may mark PP_3303 as **candidate/redundant paralog** (covered but not the primary).

FabA (PP_4174, EC 4.2.1.59 + 5.3.3.14). Bifunctional 3-hydroxydecanoyl-ACP dehydratase/isomerase — supplies both a dehydratase activity and, critically, the trans-2 → cis-3 isomerization that commits chains to the unsaturated series. Together with FabB it covers the entire *cis*-unsaturation branch (F005).

FabV (PP_4635, EC 1.3.1.9, NADH). The **sole** enoyl-ACP reductase (F002). This is a defining lineage feature: proteome searches (organism_id:160488) returned **0 hits for gene:fabI and 0 for gene:fabL**, while FabV is annotated as TER-reductase-family enoyl-ACP reductase with PATHWAY "fatty acid biosynthesis". The physiological importance of FabV in *Pseudomonas* is directly demonstrated in the sister species *P. aeruginosa* PAO1, where *fabV* deletion made cells >2,000-fold more triclosan-sensitive (PMID: 19933806). Because it is a single point of coverage, FabV warrants a **covered but non-redundant** flag.

Present but weakly annotated (covered, promote to review)

PP_4379 / FabH (EC 2.3.1.180, KAS-III). The **sole** KAS-III chain initiator: a proteome search for EC 2.3.1.180 in taxon 160488 returns exactly one hit, PP_4379, and no gene is annotated `fabH` (F003). It is, however, PE:4 "Predicted", carries **no recommended name and no EC in the current UniProt entry**, and the review metadata mislabels it "Beta-ketoacyl-ACP synthase I (EC 2.3.1.180)" — an internal inconsistency, since EC 2.3.1.180 is KAS-III (FabH), whereas KAS-I (EC 2.3.1.41) is the true *fabB* = PP_4175. Its 309-aa length is consistent with the FabH family. **Promote to full `fetch-gene` review** and re-annotate as *fabH*.

PP_1183 (Sfp/EntD-type PPTase, EC 2.7.8.7 candidate). ACP activation is an apparent gap by name — proteome searches for EC 2.7.8.7 and "holo-[acyl-carrier-protein] synthase" returned **0 hits**, and the InterPro AcpS/PPTase family IPR008278 in this proteome contains **exactly one protein: PP_1183**, an EntD/Sfp-type "Enterobactin synthase component D" (F006/F007). This mirrors the known *Pseudomonas* architecture in which a single Sfp-type PPTase activates both the primary FAS-II ACP and secondary-metabolite carrier proteins, replacing the enterobacterial two-PPTase (AcpS + EntD) system. This is a **lineage substitution, not a true gap**; promote PP_1183 to full review to confirm it acts on AcpP.

Missing from the list but essential (add)

AcpP (PP_1915, Q88LL5). The acyl carrier protein — 78 aa, PE:1 "Evidence at protein level" (the highest tier) — is present and essential but **absent from the 19 candidate genes** (F001). It sits in the canonical cluster PP_1912 *plsX* – PP_1913 *fabD* – PP_1914 *fabG* – **PP_1915 *acpP*** – PP_1916 *fabF*, mirroring the *E. coli fabD-fabG-acpP-fabF* operon. Without ACP the entire module is mechanistically inoperable; this is the single most important metadata correction.

Out-of-scope / likely over-propagated (exclude)

Gene	Locus	UniProt annotation	Why excluded
fadD-I	PP_4549	Long-chain acyl-CoA ligase, EC 6.2.1.3	PATHWAY = β-oxidation; fatty-acid **activation**, not synthesis (F004)
fadD-II	PP_4550	Long-chain acyl-CoA ligase, EC 6.2.1.3	Same as above (F004)
PP_2783	PP_2783	2,3-dihydroxy-p-cumate dehydrogenase, EC 1.3.1.58	PATHWAY = p-cumate degradation; aromatic catabolism SDR (F004)
PP_0581	PP_0581	"SDR family", no gene/EC/function (PE:3)	Generic SDR propagated by broad GO/EC mapping (F004)
PP_1852	PP_1852	"SDR family", no gene/EC/function (PE:3)	Metadata labels it "enoyl-ACP reductase EC 1.3.1.10" but UniProt gives no such evidence; enoyl step already covered by FabV (F002/F004)
PP_2540	PP_2540	"SDR family", no gene/EC/function (PE:3)	Generic SDR (F004)
PP_4529	PP_4529	"3-oxoacyl-ACP synthase", PE:4, **84 aa**	Far too short for a ~330–410 aa KAS; fragment/spurious (F007)

5. Gaps, Ambiguities, and Likely Over-Annotations

No genuine pathway gaps. All four elongation activities, initiation, loading, malonyl-CoA supply, the carrier, and the unsaturation branch are encoded. The two apparent "gaps" resolve to lineage substitutions:

- **ACP activation (EC 2.7.8.7):** no dedicated AcpS; PP_1183 (Sfp/EntD-type) is the sole PPTase and the most plausible AcpP activator (F006/F007). Mark **covered by lineage substitution**, not gap.
- **Enoyl reduction (EC 1.3.1.9):** no FabI/FabL; FabV is the sole reductase (F002). Mark **covered (single point)**.

Paralog ambiguity. - **KAS-II:** two paralogs share EC 2.3.1.179 — FabF (PP_1916) and PP_3303. Primary = FabF; PP_3303 = candidate/redundant. - **β-keto reductase family:** PP_0581, PP_2540 (and the mislabelled PP_1852) are bare SDRs pulled into the bucket by the broad SDR/oxidoreductase GO terms. FabG (PP_1914) is the real reductase; the others are **likely over-annotation**.

Nomenclature/EC mislabels (curation-relevant). - PP_4379: labelled "synthase I" but EC 2.3.1.180 = KAS-III/FabH. Fix name → *fabH*. - PP_1852: metadata claims "Enoyl-ACP reductase (NADPH, B-specific) EC 1.3.1.10" but UniProt carries only a generic SDR annotation with no EC. Do not treat as an enoyl reductase. - FabD bucket `ppu00074` and several genes bucketed to `ppu00780` reflect KEGG sub-map fragmentation; functionally all are FAS-II.

Broad EC/GO over-propagation. The recurring pattern is generic **SDR-family** and **"oxidoreductase"** annotations (PP_0581, PP_1852, PP_2540, PP_2783) being swept into a fatty-acid-synthesis bucket. These should be demoted unless direct evidence links them to acyl-ACP substrates.

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6. Mechanistic Model

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acetyl-CoA
    │  AccABCD (accA PP_1607 / accB PP_0559 / accC PP_0558 / accD PP_1996)
    ▼
malonyl-CoA
    │  FabD (PP_1913)              apo-AcpP (PP_1915)
    ▼                                  │ PP_1183 (Sfp/EntD PPTase, EC 2.7.8.7)
malonyl-ACP  ◄───── holo-AcpP  ◄───────┘
    │
    │  FabH / PP_4379 (KAS-III, EC 2.3.1.180)  + acetyl-CoA
    ▼
acetoacetyl-ACP
    │
┌───┴──────────────── ELONGATION CYCLE ─────────────────┐
│  ① FabB PP_4175 (KAS-I) / FabF PP_1916 (+PP_3303) KAS-II │
│  ② FabG PP_1914  (β-keto reduction, NADPH)               │
│  ③ FabZ PP_1602 / FabA PP_4174 (dehydration)             │
│  ④ FabV PP_4635  (enoyl reduction, NADH — SOLE ENR)      │
└───────────────────────────┬───────────────────────────┘
                            │  at C10: FabA isomerization (5.3.3.14)
                            ▼
                 cis-3-decenoyl-ACP ──FabB──► 16:1 / 18:1 (cis-vaccenate)
                            │
                            ▼
                 acyl-ACP → membrane phospholipids (PlsX/B/C, downstream)
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The narrative: KT2440 runs a textbook dissociated FAS-II with two *Pseudomonas*-specific twists — a **single Sfp/EntD-type PPTase** replacing the enterobacterial AcpS, and a **single NADH-dependent FabV** replacing FabI as the enoyl reductase (with no FabL backup). These are exactly the features a naive homology transfer from *E. coli* would miss, which is why the module needs the explicit PPTase sub-step and the FabV-only enoyl node.

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7. Module and GO-Curation Recommendations

Module-step dispositions:

Module step	Disposition	Rationale
Malonyl-CoA formation (ACC)	**covered**	accABCD canonical
Acyl carrier protein	**covered — metadata fix**	add AcpP PP_1915 (F001)
ACP activation (holo-ACP)	**covered — lineage substitution**	PP_1183 Sfp/EntD PPTase; add sub-step + note (F007)
Malonyl-ACP loading (FabD)	**covered**	PP_1913
Chain initiation (KAS-III)	**covered — candidate_uncertain annotation**	PP_4379/FabH; promote & re-annotate (F003)
Condensation KAS-I/II	**covered**	fabB, fabF (+PP_3303 redundant)
β-keto reduction	**covered**	fabG; exclude SDR paralogs
Dehydration	**covered**	fabZ (+fabA)
Enoyl reduction	**covered — single point**	FabV only; no FabI/FabL (F002)
cis-unsaturation branch	**covered**	fabA + fabB

Module-boundary / document actions: 1. **Add AcpP (PP_1915)** as an explicit module player — the generic outline names "PSEPK canonical AcpP" but the locus is absent from candidates. 2. **Add an ACP-activation (PPTase, EC 2.7.8.7) sub-step** to the module with PP_1183 as a *Pseudomonas*-specific bifunctional Sfp/EntD-type player, distinct from enterobacterial AcpS. Add a GO/module note that in *Pseudomonas* a single Sfp-type PPTase serves both primary and secondary metabolism. 3. **Flag the enoyl-reduction node as FabV-only** (NADH, EC 1.3.1.9) and remove the PP_1852 "NADPH-dependent reductase (EC 1.3.1.10)" variant from the module unless direct evidence emerges — the current outline over-represents an enoyl-ACP reductase paralog with no supporting UniProt annotation. 4. **Remove/relabel out-of-scope genes** fadD-I/II, PP_2783, PP_0581, PP_2540, PP_4529 from the FAS-II bucket. 5. **No new GO term requests are strictly required**, but a curation note distinguishing the *Pseudomonas* single-Sfp-PPTase architecture from the two-enzyme enterobacterial model would prevent future "acpS gap" false positives.

Existing generic module boundaries are largely correct; the main corrections are that the module should explicitly encode the **PPTase activation sub-step** and the **FabV-only enoyl node**, and should not carry an NADPH enoyl-reductase variant for this organism.

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8. Genes to Promote to Full `fetch-gene` Review

1. **PP_4379 (FabH / KAS-III, EC 2.3.1.180)** — sole chain initiator, PE:4, no name/EC in UniProt, mislabelled "synthase I". Highest priority: it anchors initiation and its annotation is internally inconsistent (F003).
2. **PP_1183 (Sfp/EntD-type PPTase, EC 2.7.8.7 candidate)** — sole PPTase; confirm it activates AcpP (holo-ACP synthesis) versus being dedicated to siderophore carriers (F007).
3. **AcpP (PP_1915)** — add to the module and confirm as the canonical FAS-II carrier (PE:1) (F001).
4. *(Secondary)* **PP_3303** — confirm KAS-II redundancy vs. specialized role; **FabV (PP_4635)** — confirm sole-enoyl-reductase status and cofactor.

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9. Evidence Base

Evidence	Source	Type	Supports
FabV deletion → >2,000-fold triclosan hypersensitivity in *P. aeruginosa* PAO1; FabI-only mutant stays resistant	PMID: 19933806	Direct experiment (sister species)	FabV is a functional, physiologically dominant enoyl-ACP reductase in *Pseudomonas* (F002)
acpP PP_1915 PE:1, 78 aa, in plsX-fabD-fabG-acpP-fabF cluster	UniProt Q88LL5	Protein-level annotation (target)	AcpP present & essential (F001)
0 hits for fabI/fabL; fabV = TER-reductase-family ENR	UniProt proteome 160488	Homology/ proteome census (target)	FabV sole ENR (F002)
EC 2.3.1.180 → single hit PP_4379; no fabH gene	UniProt proteome 160488	Proteome census (target)	Sole KAS-III initiator (F003)
ACC, FabD, FabG, FabZ, FabF, FabB, FabA pathway-consistent	UniProt (target)	Annotation (target)	Core elongation covered (F005)
IPR008278 PPTase family = single protein PP_1183 (EntD/ Sfp)	UniProt/ InterPro (target)	Family census (target)	ACP activation by lineage substitution (F007)
fadD-I/II PATHWAY = β-oxidation; PP_2783 = p-cumate	UniProt (target)	Annotation (target)	Out-of-scope genes (F004)

The **species-transfer strength** varies. Target-organism UniProt/InterPro evidence is **strong/ direct** for gene presence and family membership. The FabV physiological-dominance claim is **direct in *P. aeruginosa* PAO1** and transfers **strongly** to *P. putida* KT2440 (same genus, identical FabV-family enzyme, no FabI/FabL in either). The supporting quote from PMID: 19933806: *"Upon deletion of the fabV gene, the mutant strain became extremely sensitive to triclosan (>2,000-fold more sensitive than the wild-type strain), whereas the mutant strain lacking FabI remained completely resistant to the biocide."* No dedicated KT2440 biochemical study of PP_4379/FabH or PP_1183 was found; these remain **homology/family-level**

inferences (hence the promotion recommendations). Retrieved rhamnolipid/PHA papers (e.g., PMID: 41117507, PMID: 41046915, PMID: 42393666) confirm that de novo FAS supplies acyl precursors in KT2440 metabolism but do not test individual FAS-II gene functions.

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10. Limitations and Knowledge Gaps

- **No direct KT2440 biochemistry** was located for PP_4379/FabH (initiation) or PP_1183 (ACP activation); both rest on family/EC inference. The FabH annotation is internally inconsistent and PE:4.
- **PP_1183 substrate scope** is unconfirmed — whether it activates AcpP in addition to siderophore carrier proteins is inferred from the *Pseudomonas* single-PPTase paradigm, not demonstrated in KT2440.
- **PP_3303 role** (specialized vs. redundant KAS-II) is unresolved.
- The **SDR paralogs** (PP_0581, PP_2540, PP_1852) have no functional data; their exclusion is based on absence of specific annotation, which is suggestive but not definitive.
- Analysis was annotation- and literature-based (UniProt, InterPro, KEGG, PubMed); no experimental or structural validation was performed in this review.

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11. Proposed Follow-up Actions

1. **Curate AcpP (PP_1915) into the module** as the carrier player — immediate, high-value metadata fix.
2. **Promote PP_4379 to full `fetch-gene` review**; re-annotate as *fabH*, EC 2.3.1.180, KAS-III initiator; correct the "synthase I" mislabel.
3. **Promote PP_1183 to full review**; add an ACP-activation sub-step (EC 2.7.8.7) and confirm bifunctional Sfp/EntD-type activity on AcpP (e.g., in-vitro holo-ACP synthase assay, or conditional-lethality/ACP-modification phenotype).
4. **Flag FabV (PP_4635) as the sole enoyl reductase**; a Δ*fabV* triclosan-sensitivity assay in KT2440 would directly confirm (mirroring the PAO1 result).
5. **Remove out-of-scope genes** (fadD-I/II, PP_2783, PP_0581, PP_2540, PP_4529) and the metadata's PP_1852 "enoyl reductase" claim from the FAS-II bucket.

6. **Add a curation note** distinguishing the *Pseudomonas* single-Sfp-PPTase architecture from the enterobacterial AcpS+EntD system, to suppress future false "acpS gap" flags.

Key References

- Zhu L, *et al. Triclosan resistance of Pseudomonas aeruginosa PAO1 is due to FabV, a triclosan-resistant enoyl-acyl carrier protein reductase.* PMID: 19933806
- *Heterologous rhamnolipid biosynthesis — role of de novo fatty acid synthesis in supplying the hydrophobic moiety.* PMID: 41117507
- *Co-feeding phenolic acids with fatty acids for PHA synthesis in P. putida KT2440.* PMID: 41046915
- *Acetate as alternative carbon source for rhamnolipid production in P. putida KT2440 (de novo FAS precursor supply).* PMID: 42393666
- Primary annotation sources: UniProt proteome UP000000556 / taxon 160488 (Q88LL5 AcpP, Q88E33 FabV, Q88ES4 PP_4379, Q88NM4 PP_1183, and core fab/acc entries); InterPro IPR008278 (PPTase family); KEGG ppu00061.
